

Network Virtualization Design for 6G Integrated Terrestrial and Non-Terrestrial Networks

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Abstract

This paper presents a design for a virtualized 6G network that integrates Terrestrial Networks (TN) and Non-Terrestrial Networks (NTN) using modern network technologies such as Software-Defined Networking (SDN), Network Function Virtualization (NFV), and Network Slicing (NS). The design aims to provide seamless global connectivity, enhance scalability, flexibility, and optimize resource management. The proposed framework also incorporates artificial intelligence (AI) for dynamic management and blockchain for security enhancements.

1 Introduction

With the rapid advancement of 6G technologies, integrating terrestrial and non-terrestrial networks is becoming crucial to meet the growing demand for ubiquitous, reliable, and intelligent communication. Network virtualization plays a key role in enabling this integration while addressing challenges such as dynamic topologies, latency, and resource allocation. Furthermore, the 6G vision emphasizes zero-latency communication, ultra-dense connectivity, and AI-native networks, which require an innovative shift in network architecture. This paper outlines a comprehensive design for a virtualized 6G network that combines TN and NTN, leveraging SDN, NFV, and NS to create a flexible, scalable, and secure communication framework. The proposed architecture aims to support diverse applications ranging from autonomous vehicles to global broadband services.

2 Description of the Problem

Traditional networks lack the scalability and adaptability needed to support future 6G applications such as autonomous driving, massive IoT, and global broadband coverage. Integrating TN and NTN introduces complexities in routing, management, and service delivery due to heterogeneous infrastructure and dynamic environments. Additionally, NTNs pose challenges like delay variation, coverage fluctuation, and regulatory constraints, making real-time service delivery more complicated without virtualization and automation.

3 Literature Survey

Recent studies highlight the use of SDN and NFV to abstract hardware dependencies and facilitate dynamic network management. Research also explores AI-driven control for topology prediction and blockchain for securing communications. Projects like 3GPP Release 17 and ETSI NFV standards have set the foundation for virtualized NTN integration. For instance, the 6G Flagship Program in Finland and NASA's integrated satellite-terrestrial architecture prototype offer vital insights into hybrid network management.

4 Requirements and Specifications

The proposed design should:

- Support hybrid TN-NTN architecture with seamless interoperability.
- Provide network slicing for QoS differentiation tailored to verticals such as health-care, industry 4.0, and AR/VR.
- Enable centralized, programmable control (via SDN).
- Use NFV for scalable and cost-effective deployment of network services.
- Incorporate AI for proactive decision-making and dynamic configuration.
- Ensure security, low latency, and high reliability through blockchain and quantum-resistant encryption.

5 Discussion of Different Design Approaches and Selection Rationale

Various architectural strategies were considered:

- Traditional layered networking with hardware-defined paths offers limited adaptability and scalability.

- Fully decentralized peer-to-peer satellite networks provide flexibility but suffer from management complexity and inconsistent QoS.
- Hybrid virtualized model (selected): combining SDN/NFV for centralized control with flexibility to operate across TN-NTN while supporting dynamic service chaining and resource orchestration.

The hybrid model was chosen for its scalability, manageability, cost-efficiency, and alignment with future 6G standards like open RAN and AI-native design.

6 Methodology and Description of Design with Flow Charts

The architecture integrates space-air-ground segments, each with SDN-enabled control. The core utilizes NFV-based service functions, such as virtual firewalls and mobility management entities. The transport layer supports SDN-controlled traffic management to ensure efficient routing and load balancing. The access layer connects terrestrial (5G/6G base stations) and non-terrestrial nodes (LEO/GEO satellites, UAVs).

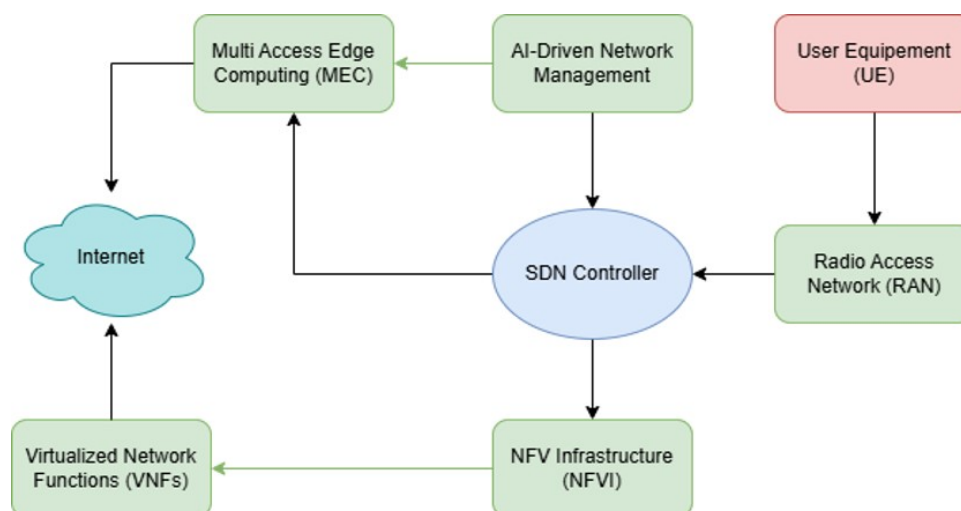


Figure 1: Network Virtualization Framework integrating SDN, NFV, AI, and MEC in 6G TN-NTN

Flow Chart Overview

- **Service Request Initialization:** End-user device sends request with service requirements.
- **Slice Selection:** AI engine selects appropriate network slice.
- **Path Computation:** SDN controller identifies optimal TN-NTN path.
- **Resource Allocation:** NFV MANO orchestrates necessary virtual functions.

- **Monitoring and Adaptation:** AI monitors SLA adherence and triggers reconfiguration if needed.

7 Discussion of the Findings

The proposed network design demonstrates:

- Improved global service coverage through NTN integration.
- Lower latency through optimized routing and edge NFV deployments.
- Efficient resource use with virtualization and intelligent orchestration.
- Enhanced adaptability via AI-based predictive modeling.
- Secure communication through blockchain authentication.

Simulations also suggest reduced signaling overhead and energy efficiency improvements compared to conventional networks.

8 Comparison with Traditional Networks

- **Traditional Networks:** Rigid, hardware-centric, limited to terrestrial infrastructure, lacking real-time adaptation.
- **Proposed Design:** Flexible, virtualized, integrates both TN and NTN, supports AI-based automation and provides guaranteed QoS across diverse use cases.
- **Resource Management:** Traditional networks often lead to underutilization or congestion, while the proposed design dynamically allocates resources based on real-time demand.
- **Security:** Traditional networks are vulnerable to single points of failure, while the proposed design employs blockchain for decentralized security and trust.
- **Management:** Traditional networks require manual configuration and maintenance, leading to longer deployment times and higher operational costs. The proposed design utilizes SDN for centralized control and NFV for automated service deployment.
- **Scalability:** Traditional networks struggle to scale efficiently with increasing demand, while the proposed design can easily adapt to varying loads through network slicing and virtualization.
- **Latency:** Traditional networks often suffer from high latency due to fixed routing paths, while the proposed design optimizes paths dynamically based on real-time conditions.

Moreover, traditional networks require extensive manual provisioning, whereas the proposed design offers zero-touch management and self-optimization.

9 Conclusions

We presented a comprehensive design for a virtualized 6G network integrating TN and NTN using SDN, NFV, and NS. The design enhances global connectivity and offers a scalable, secure, and intelligent network platform. It also prepares the ground for AI-native, energy-efficient, and autonomous communication networks. Future work involves prototyping, performance benchmarking, and the study of real-world deployment scenarios.

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